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Upper limb exoskeletons for assistance in activities of daily living (ADLs)

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CONTENTS

Summary

Introduction 6

Chapter I – General Overview of Upper Limb Exoskeletons 9

1.1 Historical Development of Upper Limb Exoskeletons 9

1.1.1 Birth of Upper Limb Exoskeletons for Assistance in Activities of Daily Living 9

1.2 General Classification of Upper Limb Exoskeletons 9

1.2.1 Upper Limb Exoskeletons for Human Power Augmentation 10

1.2.1.1 EksoVest 10

1.2.1.2 Robo-Mate Exoskeleton 11

1.2.2 Upper limb Exoskeletons for Medical Rehabilitation 11

1.2.2.1 Parallel Actuated Shoulder Exoskeleton 11

1.2.2.2 NEUROExos 12

1.2.3 Assistive Upper Limb Exoskeletons 13

1.2.3.1 ASSIST 13

1.2.3.2 KULEX 13

1.2.3.3 Vertical Motion Support Device 14

1.2.3.4 MyoPro 14

1.2.3.5 Hybrid Assistive Limb (HAL) 15

1.2.3.6 A Low-Cost Upper Limb Exoskeleton 15

1.2.3.7 SPexo 15

Existing Active Upper Limb Exoskeletons for Assistive purposes for elderly and weak people 16

Chapter II - Design Principles and Manufacturing of Upper Limb Exoskeletons 17

2.1 Upper Limb Anatomy 17

2.2 Mechanical Design of an Assistive Upper Limb Exoskeleton 19

2.3 Design Difficulties 21

2.4 Materials of Assistive Upper Limb Exoskeletons 21

Chapter III - Control Strategies for Assistive Upper Limb Exoskeletons 22

3.1 Assistive mode 23

3.1.1 Passive Control 23

3.1.1.1 Passive Trajectory Tracking 23

3.1.1.2 Passive Mirroring 23

3.1.1.3 Passive Stretching 23

3.1.2 Triggered Passive Control 24
3.1.3 Partially Assistive Control 24
3.1.3.1 Impedance/Admittance Control 24
3.1.3.2 Attractive Force-Field Control 24
3.1.3.3 Model-Based Assistance 25
3.1.3.4 Offline Adaptive Control 25
3.2 Corrective mode 25
3.2.1 Tunneling 25
3.2.2 Coordination Control 25
Chapter IV - Control Systems for Assistive Upper Limb Exoskeletons 26
4.1 Control Methodologies 26
4.1.1 Control Methods based on Input Signals to the Controller 26
4.1.1.1 Control methods based on Human Biological Signals 27
4.1.1.2 Control methods based on Non-Biological Signals 27
4.1.1.3 Platform Independent control methods 27
Chapter V - Actuation, Power Transmission and Sensing Methods 28
5.1 Actuation 28
5.1.1 Electric Motors 28
5.1.2 Pneumatic Actuators 28
5.1.3 Hydraulic Actuators 28
5.1.4 Series Elastic Actuators (SEAs) 29
5.1.5 Pneumatic Artificial Muscle Actuators (PAMAs) 29
5.2 Power Transmission 29
5.2.1 Direct drive, gear or linkage 29
5.2.2 Cable-driven 29
5.3 Sensing and Estimation 30
5.3.1 Classification of the Sensors used in Upper Limb Robotic Exoskeletons 30
5.3.1.1 Position sensors 30
5.3.1.2 Force/Torque sensors 31
5.3.1.3 IMU sensors 31
5.3.1.4 EMG sensors 31
5.3.1.5 EEG sensors 31
A Comparison between different Assistive Exoskeletons in terms of Sensing Methods and Applications 32
CHAPTER VI- Possible Challenges in Upper Limb Exoskeleton Development 33
6.1 Kinematic compatibility 33

6.2 Workspace limitation 33
6.3 Singularity problem of mechanical system 33
6.4 Discomfort and Misalignment 33
6.5 Human-Robot interaction 34
CHAPTER VII – Evaluation methods, clinical overview and market integration of upper limb assistive exoskeletons 35
7.1 Performance Evaluation Methods 35
7.2 Clinical Reviews of Various Assistive Robotic Exoskeletons 35
7.3 Market Integration and Economic Impact 36
CHAPTER VIII - Discussion and Future Prospective 37
8.1 Key Innovations and Developments 37
8.2 Barriers to Progress 38
CHAPTER IX – Conclusions 39
BIBLIOGRAPHY 41
LIST OF FIGURES/TABLES: 45

Introduction

With advancements in technology and medicine, during the 21st century, the global percentage of the elderly population has started to rise rapidly. According to a report from the World Health Organization (WHO), nowadays most individuals can expect to live beyond the age of 60 [1]. The United Nations estimates that by 2050, the number of older adults will reach approximately 2 billion [2]. Additionally, it is expected that developing nations will be the house of a greater number of elderly individuals compared to the developed countries [3]. Figure 1 illustrates the estimates and projections for the increase in elderly citizens across developed and developing regions. The aging process leads to a decline in skeletal and muscle structures, which cause a reduction in both muscle strength and volume in older adults, therefore elderly people will be more susceptible to various age-related diseases. On average, older individuals lose about 3% of their muscle strength each year [4]. Research shows that the aging process also affects movements involving the upper limbs in the elderly. This decline in muscle power in the upper limbs can reduce mobility and restrict participation in daily activities. Approximately 15% of individuals over the age of 65 require assistance with everyday tasks [5].

However, aging is not the only cause which can lead to upper limbs disability, in fact a lot of people can be affected by deabilitating diseases also during their young age. For example, spinal cord injury (SCI) and stroke are among the leading causes of disability worldwide. In the United States alone, approximately 795,000 individuals experience a new or recurrent stroke each year [6] and between 232,000 and 316,000 people are estimated to live with SCI [7]. Roughly one-third of stroke survivors are left with significant disabilities [8], with impaired upper limb function being one of the most common motor coordination issues following a stroke. It is widely known that physically weak individuals with disabled limbs can perform certain tasks in rehabilitation settings, where they can train to improve their skills, however, in most cases, they struggle to transfer these skills to activities of daily living (ADLs) [9].

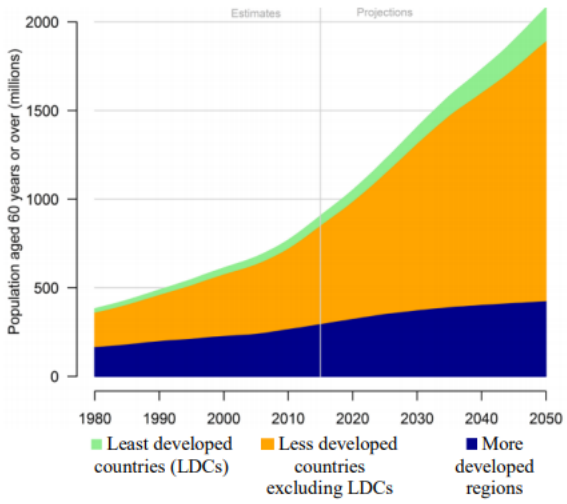


Figure 1. Estimation and projection of elderly people of the age of 60 years and beyond from 1980 to 2050.

To support people with upper limb disabilities, due to both advanced age or diseases, **Assistive Upper Limb Exoskeletons** (Figure 2) **have been invented**. Robotic assistive exoskeletons for the upper limbs can help people in performing ADLs such as: unlocking a door, combing their hair, buttoning clothes, eating and drinking. Activities like eating and drinking frequently require assistance from caregivers. However, relying on these people presents challenges, including 1) a shortage of caregivers due to a declining birth rate and a rapidly aging population and 2) the negative psychological impact on individuals who dislike feeling dependent or treated like children, and this represents an additional reason of why such rehabilitation systems have been introduced on the market. **In particular, the purpose of this thesis is to explain the state of the art of upper limb exoskeletons for assistance in ADLs**, which are also known as **Assistive Exoskeleton Devices**. The term assistive to which this current paper is targeted means that the device can provide additional torque to help disabled or elderly people in completing various physical motions.

Upper limb exoskeletons are wearable robotic systems consisting of joints and links, designed to assist, support or enhance the movement and function of the upper body, specifically the arms, shoulders and wrists. These devices are typically anthropomorphic, meaning they are designed to closely mimic the structure and movement of the human upper limb musculoskeletal system.

The debate surrounding upper limb exoskeletons has gained importance recently due to advancements in technology and their broader applications. In general, upper limb exoskeletons are not only used in the **healthcare sector** for assisting disabled or weak patients, but they are also employed in **industrial sectors**, and **their introduction was primarily driven by the need to enhance physical performance and address specific challenges in both fields**.

In healthcare, these devices are used for rehabilitation, particularly for stroke survivors, individuals with neuromuscular disorders and physically weak elderly people. They assist in restoring motor function by enabling repetitive and controlled movements that are crucial for recovery. In the industrial sector, exoskeletons were introduced to reduce the risk of musculoskeletal injuries caused by repetitive tasks or heavy lifting. By supporting workers' upper limbs, these devices reduce strain on muscles and joints, which helps to prevent workplace injuries. Moreover, they enhance productivity by allowing workers to handle tasks with less effort and fatigue, boosting the overall efficiency.

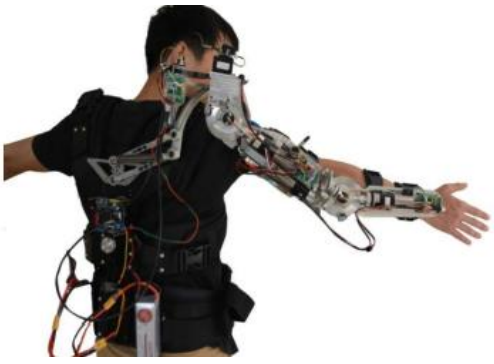


Figure 2. Prototype of an upper limb exoskeleton for assistance purposes in ADLs.

However, there are several concerns fueling the debate:

a. ***Ergonomics and Long-Term Effects***: While upper limb exoskeletons can reduce muscle tension in specific tasks, questions remain about their long-term effects on muscle deconditioning or overloading of other parts of the body due to the load redistribution. In some cases, discomfort or unintended consequences for users have been reported, raising concerns especially about their use in the industrial field.

b. ***Technological Limitations***: Upper limb exoskeletons are still being refined in terms of user comfort, flexibility and compatibility with human movements. Current devices can be cumbersome or fail to naturally adapt to users' needs. This ongoing development makes it difficult to integrate these devices seamlessly into everyday activities or workplaces.

c. ***Cost and Accessibility***: Exoskeletons, particularly in healthcare, are expensive to develop and implement, making them less accessible to use. In industrial sectors, cost-benefit ratios remain a topic of debate as companies weigh the financial expense of exoskeletons versus the benefits of injury prevention.

This thesis is composed of **11 chapters**: In the first chapter a general overview of upper limb exoskeletons is given, discussing their origin and classification methods. Chapter II focuses on the design principles and difficulties of these systems, moreover, the mechanical design of a particular upper limb exoskeleton for assistance in ADLs and materials of assistive upper limb exoskeletons are discussed. Chapter III explains the control strategies exploited for assistive upper limb exoskeletons, while in Chapter IV the main control methodologies to regulate and supervise these systems are presented. Chapter V aims to explain how upper limb exoskeletons work, focusing on actuation methods, power transmission and sensing methods and the Chapter VI discusses current challenges in the design and use of upper limb exoskeletons. Moving forward, we have the Chapter VII which explains how performances of these systems are evaluated, while in Chapter VIII different assistive upper limb exoskeletons are discussed. Chapter IX aims to explain the economic impact of these systems and how they have been introduced into the market, while Chapter X is focused on both current and future innovative discoveries about upper limb exoskeletons, discussing also major problems with this technology. Finally, in Chapter XI, conclusions are drawn.

More precisely, this paper aims to provide a general overview of exoskeletons for the upper limbs, dealing with all the most important aspects, however particular attention will be paid to those used for assistance in performing ADLs which will be discussed in more depth in each single chapter.

Chapter I – General Overview of Upper Limb Exoskeletons

1.1 Historical Development of Upper Limb Exoskeletons

The concept of the exoskeleton dates back to the late 19th century, but the first successful prototype, known as Hardiman, was created in the 1960s [10] by General Electric and the U.S. military. Hardiman was developed primarily for military use to increase soldiers' strength and performance. It utilized a master-slave control system with multiple hydraulic actuators, where two interconnected exoskeletons were involved [11]. The inner exoskeleton would mimic human movement, which then trigger the outer exoskeleton's actuator. However, Hardiman never moved beyond the prototype phase due to its weight and complexity. Later, Kazerooni et al. [12] introduced an upper-limb exoskeleton with the concept of physical human-robot interaction (**pHRI**, *see paragraph 6.5*). Unlike the master-slave system, pHRI enables direct mechanical power transfer between human and robot. In recent decades, upper-limb exoskeletons for rehabilitation and service purposes have gained significant attention from the biomedical and engineering communities, as they hold great promise for improving the lives of individuals with physical impairments.

1.1.1 Birth of Upper Limb Exoskeletons for Assistance in Activities of Daily Living

In 1995, Homma and Arai introduced one of the earliest concepts of upper-limb assistive device with a parallel mechanism aimed at aiding individuals with disabilities or muscle weakness in performing daily activities [13]. The design, depicted in Figure 3, consists in attaching strings to the human hand, with arm movements being facilitated by adjusting the length of each string. The motor and control systems were placed behind the wheelchair. However, this design remained conceptual and was never developed into a physical prototype [14].

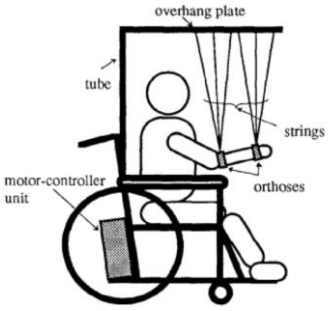


Figure 3. One of the earliest upper limb assistive device concept by Homma and Arai.

1.2 General Classification of Upper Limb Exoskeletons

Upper limb exoskeletons are designed to operate in parallel with the human arm and are attached at various points along the limb. As a result, the robot must be capable to accommodate different arm lengths [11]. In reality, there is no univocal categorization in the literature regarding exoskeletons for the upper limbs; however, a general classification is reported in Figure 4 where upper limb exoskeletons are classified based on degrees of freedom, actuation type, supported joints and applications.

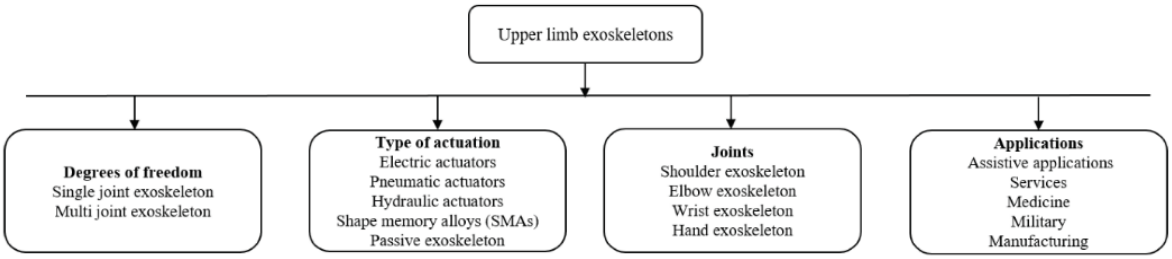


Figure 4. General classification of upper-limb exoskeletons.

In particular, considering their field of application, experts have classified upper limb exoskeletons into three main categories: **exoskeletons for human power augmentation (motion assistance exoskeletons), exoskeletons for medical rehabilitation** and **assistive exoskeletons**.

1.2.1 Upper Limb Exoskeletons for Human Power Augmentation

These exoskeletons are designed to assist users by amplifying or enhancing their movements and they are often employed to reduce worker fatigue in very harsh industrial [11]. In recent years, numerous new systems have been developed and tested. However, only a small number of these systems have been successfully commercialized, while many others are still in the research phase.

1.2.1.1 EksoVest

Ekso-BIONICS introduced a **passive** - *meaning that it does not exploit external actuators in order to enhance the user's strength, but instead it employs spring or dampers (see paragraph 5.1)* - upper-body exoskeleton called EksoVest, as illustrated in Figure 5. EksoVest utilizes two distinct mechanisms, known as the "moment generation" and "hinge mechanism", to reduce the load and fatigue experienced by workers on the shop floor. This exoskeleton provides partial support for the weight carried by the user, particularly during overhead tasks. Kim et al. [15] has evaluated the performance of EksoVest in a simulated overhead work environment. Their findings showed that the vest decreased shoulder abduction range of motion (ROM) by 10% and increased the center of pressure velocity in the front-to-back (anteroposterior) direction by 12% [15]. Additionally, it significantly reduced spine loading, especially during overhead drilling activities.



Figure 5. EksoVest by Ekso Bionics.

1.2.1.2 Robo-Mate Exoskeleton

Stadler et al. [16] created Robo-Mate to amplify power in industrial settings, as illustrated in Figure 6, where the device aids users in lifting and moving various objects over short distances. The mechanical structure of this exoskeleton is modular, supporting different levels of activity. It consists of three main parts: a trunk module, a passive arm and an active arm. The trunk module is specifically designed to protect the user's lower back from excessive compression forces. The passive arm features a four-bar spring-loaded parallelogram structure that provides gravity compensation. This arm includes an adjustable mechanism connected with one end of the spring inside the parallelogram, allowing users to modify the level of assistance. Thanks to Robo-Mate's modular design, users can switch between passive and active assistance modes based on their needs by swapping the spring with a wire. Additionally, a brushless DC motor with a worm gear drives the cable and it is positioned at the trunk end of the module. It is made by aluminium and the entire assembly weighs around 2.3 kg, not including the battery pack. Safety and risk assessments were also integrated into the design, making Robo-Mate sufficiently robust to be used in the automotive sector.



Figure 6. Robo-Mate.

1.2.2 Upper limb Exoskeletons for Medical Rehabilitation

This category of exoskeletons is designed for the rehabilitation of patients with physical impairments, such as post-stroke paralysis, amyotrophic lateral sclerosis or other motor and cognitive disabilities [11]. Exoskeletons belonging to this category are either mounted on a mobile/stationary platform and can not be carried by the users independently or can be used only in the clinical settings under appropriate supervision. Key challenges for these exoskeletons include ensuring compliant actuation and selecting an optimal control algorithm.

1.2.2.1 Parallel Actuated Shoulder Exoskeleton

Hsieh et al. [17] introduced a parallel-actuated shoulder mechanism, as shown in Figure 7, which consists of two spherical mechanisms: two slider crank mechanisms and a gravity-balancing system. The actuators are positioned side by side in order to enhance the system's inertia characteristics while maintaining a compact and lightweight design. To address potential misalignment between the exoskeleton and the user's limb due to varying physiological factors of patients, an adaptive mechanism with a passive joint was incorporated. Additionally, linear series elastic actuators (SEA) were employed to obtain precise force and impedance control at the exoskeleton-limb interface. This exoskeleton is designed to be mounted on a stationary platform, which limits its versatility for other applications. Furthermore, the compatibility of the design needs to be evaluated in relation to shoulder glenohumeral (GH) movements, as these can impact the range of motion (ROM) of the user's upper limb.



Figure 7. Parallel actuated shoulder exoskeleton.

1.2.2.2 NEUROExos

Lenzi et al. [18] proposed a portable robotic elbow exoskeleton named NEUROExos, depicted in Figure 8, designed for the rehabilitation of a typical motor disorder called spasticity. The initial development of NEUROExos focused on solving two critical issues: determining the precise interaction points between the exoskeleton and the human arm and ensuring proper distribution of these points. The first challenge of localization was resolved by incorporating an adaptive mechanism for the elbow joint, consisting of two main assemblies: a double-shell link-based structure and a 4-DOF adjustable passive mechanism. The 4-DOF adjustable mechanism maintains alignment between the human elbow joint's axis and the exoskeleton, while the double-shell was used to fix the human arm with the exoskeleton structure. NEUROExos is powered by a Bowden cable-based transmission unit located remotely. In a later version, bio-inspired design features were added by introducing compliance through nonlinear elastic components and custom torsional springs. NEUROExos development was based on detailed analysis of neuroscientific motion requirements and later it was used in several experimental studies to treat neurological disorders.



Figure 8. NEUROExos.

1.2.3 Assistive Upper Limb Exoskeletons

Assistive exoskeletons are designed to assist older people in improving their ability to perform daily activities and in most cases, they have a dual function, being also used for medical rehabilitation purposes. This additional feature allows such exoskeletons to be used also for medical purposes for patients with various type of diseases (e.g.; post-stroke patients). For this reason, **exoskeletons for assistance in performing ADLs** are assistive devices and, in certain cases, they can also be used as rehabilitation devices.
Currently, a large variety of upper limb exoskeletons are available to support elderly individuals in performing ADLs [14]. In the following paragraphs different assistive upper limb exoskeletons will be briefly introduced (considering also systems that incorporate both assistive and rehabilitation purposes), but those which possess only single therapeutic rehabilitation or physical augmenting purposes (the ones which are used by workers to carry or lift heavy objects) will not be discussed.

1.2.3.1 ASSIST

One example of upper limb assistive device, called **ASSIST**, was proposed by Sasaki et al. [20]. This lightweight device, which presents only 1 DOF, assists users in wrist flexion and extension. Its effectiveness has been demonstrated through different experiments, showing that it can reduce muscle fatigue and provide additional torque to help completing specific motion trajectories.

1.2.3.2 KULEX

In another study, Kim et al. [21] designed an upper limb mechanism, shown in Figure 9, aimed at assisting old individuals during ADLs. This mechanism, called **KULEX**, enhances wheelchair usability for elderly people with disabilities by integrating a self-active cast, which employs a low-energy driving system to translate residual motor functions to the upper limbs. The control unit relies on pressure and electromyography (EMG) sensors (*see paragraph 5.3.1.4*) to establish sensory feedback. The proposed system is composed of three sections: a 1-DOF hand component to assist with grasping, a 6-DOF serial robot to support forearm movement and a 3-DOF parallel mechanism to support wrist motion. Additionally, the entire structure was designed to be lightweight and energy-efficient, offering adequate maneuverability.

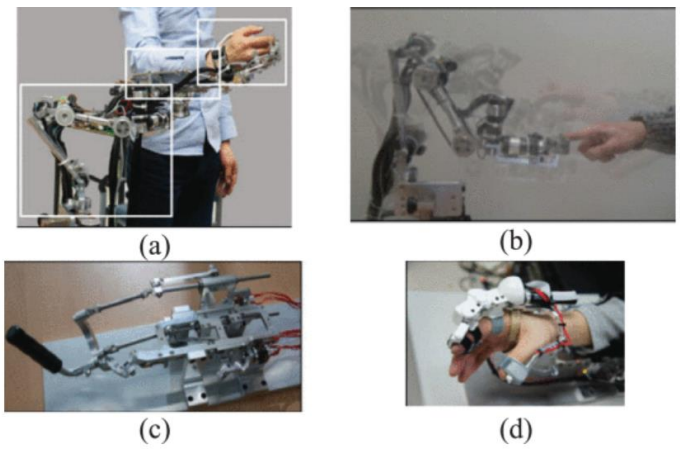


Figure 9. KULEX: a) overall system of 10 DOF, b) forearm support part, c) 3 DOF wrist part and d) grasping part.

1.2.3.3 Vertical Motion Support Device

Gu et al. [22] proposed a mechanical device, illustrated in Figure 10, to help elderly people maintaining their lifestyle independence. This system supports users in tasks like self-feeding, working on a computer and drawing. A DC motor powers a parallel mechanism via strings, while force sensors monitor assistive torque applied to the user's arm. Tactile sensors track muscle stiffness, ensuring the device's reliability, safety and usage. These stiffness sensors are easy to attach and they are unaffected by skin condition, allowing them to measure muscle contractions even through clothing. The device is primarily made of aluminum alloy.

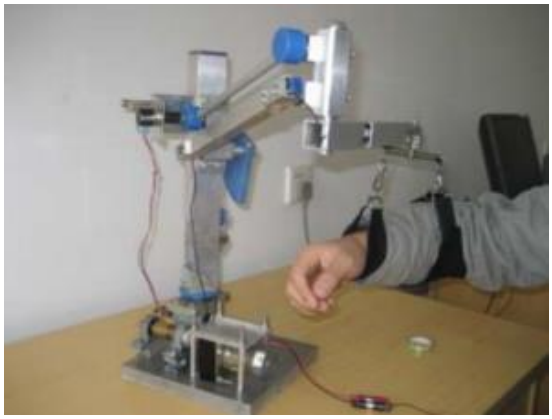


Figure 10. Assistive device to support vertical motion.

1.2.3.4 MyoPro

Myomo Inc. developed several robotic upper limb exoskeletons, including the **MyoPro for Veterans**, shown in Figure 11. This exoskeleton is commercially available and it is used to assist elderly individuals in ADLs such as woodworking, cutting food and meal preparation. The system uses EMG sensors to detect muscle contractions, helping to realize elbow flexion/extension movement [14].



Figure 11. Exoskeleton for upper limb assistance of elderly: MyoPro exoskeleton for Veterans.

1.2.3.5 Hybrid Assistive Limb (HAL)

Cyberdyne Inc. created the **HAL exoskeleton** (Figure 12). It has a single-joint prototype that could be used to assist or even train upper limb movements and it uses brain nerve to send different signals to muscles in order to control the exoskeleton [23]. Moreover, it utilizes surface EMG (sEMG) signals (*see paragraph 5.3.1.4*) to actuate the movements.

1.2.3.6 A Low-Cost Upper Limb Exoskeleton

Mohamed et al. [24] designed a low-cost 2 DOF upper limb device. This device offers different usage configurations: it can be worn as a portable exoskeleton (Figure 13) or it can be attached to a cart to reduce the load on the user. The system uses Inertial Measurement Unit (**IMU**) sensors (*see paragraph 5.3.1.3*) to measure user's arm joint angles and a potentiometer for robot angle measurements. A Worm Drive motor is used for the actuation and a PID control is implemented for precise movement.

1.2.3.7 SPexo

Lastly, Latt et al. [9] introduced a 7-DOF exoskeleton called **SPexo,** illustrated in Figure 14, that supports elderly individuals in performing daily tasks such as drinking, eating and combing hair. In addition to its assistive functions, this exoskeleton can also be used for rehabilitation purposes. The system operates using DC motors for actuation. A key aspect of its design is the incorporation of telescopic components, which allow the exoskeleton to be adjusted for users with varying body sizes. To enhance usability, the exoskeleton employs Microsoft Kinect to detect the location of objects, such as a food plate, guiding the exoskeleton toward the target position.



Figure 12. HAL for Living Support.



Figure 13. Low-cost exoskeleton portable mode.

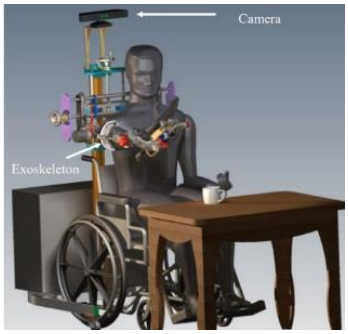


Figure 14. SPexo (CAD design).

The above listed upper limb exoskeletons are only a part of all the exoskeletons used to assist people in their daily activities. Table 1 shows a classification of the vast majority of existing **active** upper limb exoskeletons which are intended to help older people in the completion of their ADLs. The exoskeleton is said to be "active" when it uses external actuators for operation, improving the user's capabilities and skills (*see paragraph 5.1*).

Existing Active Upper Limb Exoskeletons for Assistive purposes for elderly and weak people

№	Name	Application Domain	Degrees-of-Freedom	Actuation Type	Control Strategy	Sensors
1	Active A-gear (2016) [70]	Support arm function for adults with Duchenne muscular dystrophy (DMD) in ADL (Laboratory prototype)	5- DoF	Brushless DC motor together with planetary gearboxes	Admittance control with force based interface	6 DoF force/torque sensor, hall sensors
2	Power-Assist Exoskeleton (2012) [71]	Assist physically weak people in daily life movements (fixed on wheelchair) (Laboratory prototype)	7- DoF	7 DC motors	Electromyograph (EMG)-based impedance control	Encoders and potentiometers, force/torque sensors
3	CABexo (2017) [47]	Assist elderly people and disabled people in ADL (Software model stage)	6- DoF	DC motors (cable-driven systems)	Not introduced yet	Not introduced yet
4	Serial-Parallel Exoskeleton (2011) [72]	Assist Elderly/ Physically Disabled people in ADL, mounted on wheelchair, (Laboratory prototype)	7- DoF	The Kollmorgen brushless DC motor, Faulhaber motors	Admittance control	Force/torque sensors
5	Low Cost Upper-Limb Exo [46]	Assist elderly people (Laboratory prototype)	2- DoF	Worm Gear Motor ZD1735	PID controller	IMU sensors
6	Upper-Limb Powered Exoskeleton (2007) [73] [74]	Assistive/Therapeutic for ADL (Laboratory prototype)	7- DoF	DC brushed motors	Neural-controlled (Myoprocessors)	Potentiometer, shaft encoder,Force/torque sensors
7	Portable Assistive Arm Exoskeleton (2012) [75]	Assistance/Stroke Rehabilitation (Laboratory prototype)	1- DoF	DC motors, planetary gear motors	Brain computer interface control	EEG, EMG encoders and potentiometers
8	Myosignal-Based Powered Exoskeleton (2001) [65]	Assistive, focus on the development of HMI control (Laboratory prototype)	1- DoF	DC servo motor with a planetary gearbox	Moment-based control system with embedded myoprocessor	Encoders and potentiometers, EMG sensor
9	Gopura and Kiguchi [76]	Power assist during ADL for physically weak people	7- DoF	DC servo motors	EMG – based impedance control method	Force/Torque sensors, potentiometers, encoders
10	MULOS [77]	Assist people with limb weakness or disability, attached on wheelchair (Laboratory prototype)	5-DoF	Electric motor and cable drives	PID control	Encoders
11	IKerlan's Othosis [78]	Augment physical performance of daily tasks, wheelchair mountable (Laboratory prototype)	5-DoF	DC motors and Festo pneumatic muscles	PID control	Festo torque sensors
12	ETS-MARSE [79]	Assist in daily upper limb movements and rehabilitation (Laboratory prototype)	7-DOF	Brushless DC motors	"Sliding Mode Control" Technique and computed torque control	Potentiometers
13	Naidu [80]	Assist disabled and weak people in ADL (Software model)	7 DOF	DC motors	Feedback control	Encoder
14	Sasaki [36]	Motion assist	1-DoF (Wrist)	McKibben type pneumatic muscle	Controled by Pressure control system	Encoders and potentiometers
15	Kobayashi & Hiramatsu [81]	Provide muscular support	6-DOF	McKibben type pneumatic muscle	Open loop control	Electropneumatic regulator (controls output of compressed air)
16	Ramos [82]	Complete task of lifting a payload	2 Passive – 1	Pneumatic Artificial Muscles (Hill-type muscle)	Control based on the method of GA (Genetic Algorithms) evaluate the chromosomes.	sEMG electrodes and potentiometers
17	RUPERT [83, 84]	Assistive/Rehabilitative	5-DOF Shoulder (2 -DoF, Wrist-1DoF, Elbow - 2DoF	Pneumatic muscle actuators and FES stimulators	"Iterative learning controller-based feedforward controller and Closed-loop controller with a PID-based feedback controller"	–

Table 1.

Chapter II - Design Principles and Manufacturing of Upper Limb Exoskeletons

2.1 Upper Limb Anatomy

The purpose of an exoskeleton is to mimic the kinematics and dynamics of the human musculoskeletal system, supporting limbs' motion which nowadays has become a difficult task due to the current mechanisms and actuation techniques [11]. In fact, there is not an unanimous kinematic model in biomechanics which can help us in creating upper limb exoskeletons. Additionally, the design parameters of an exoskeleton are heavily influenced by the specific application it is targeted for. Therefore, analyzing the anatomy of the human upper limb is essential for developing an exoskeleton tailored to the needs of its users.

The human upper limb has a complex skeletal structure, as shown in Figure 15A, which includes the shoulder complex, elbow complex, wrist joint and fingers. The shoulder itself is composed of four joints (the glenohumeral, acromioclavicular, sternoclavicular and scapulothoracic joints) formed between three bones: the clavicle, scapula and humerus. The glenohumeral joint, commonly referred as a *ball-socket joint*, is created by the connection of the humeral head and the glenoid cavity. Most research has focused solely on the glenohumeral joint to design a three-DOF mechanism. However, the glenohumeral joint has an instantaneous center of rotation (COR) that shifts as the human upper limb moves. Therefore, it is important to account for this dynamic center of rotation when modeling the shoulder mechanism in exoskeletons. The primary movements of the shoulder complex include shoulder abduction/adduction, shoulder internal/external rotation and shoulder flexion/extension.

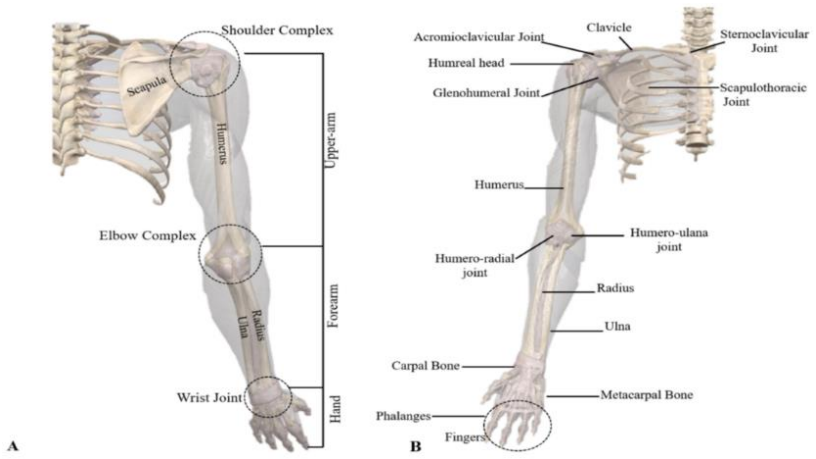


Figure 15. Anatomy of the upper limb: (A) upper limb segments and (B) shoulder, elbow and wrist skeletal structure.

The elbow joint is a synovial compound joint consisting of the humeroradial and humeroulnar joints, as shown in Figure 15B. The humeroradial joint, often described as a ball-socket joint, is formed by the connection between the humerus in the upper arm and the radius in the forearm. However, its similarity to the humeroulnar joint and the proximal radioulnar articulation restricts its movement from 3-DOF to 2-DOF. Generally, the elbow allows for forearm flexion/extension and supination/pronation. Many exoskeletons found in the literature have modeled the elbow joint with just 1-DOF for flexion and extension.

In the end, the wrist connects the forearm to the hand and consists of eight carpal bones, supported by soft tissues that reinforce the joint, as depicted in Figure 15B. The wrist possesses 2-DOF movements: wrist flexion/extension and radial/ulnar deviation. Movements of the wrist occur around an instantaneous center of rotation; however, the range of motion for this center is minimal and can be often ignored. The axes of rotation for flexion/extension and ulnar/radial deviation are typically considered to be coincident. Several exoskeletons have been designed to assist the wrist joint based on the assumption that wrist movements are generated about a fixed COR.

Overall, the arm has **seven DOF**. Figure 16 depicts various upper extremity motions.

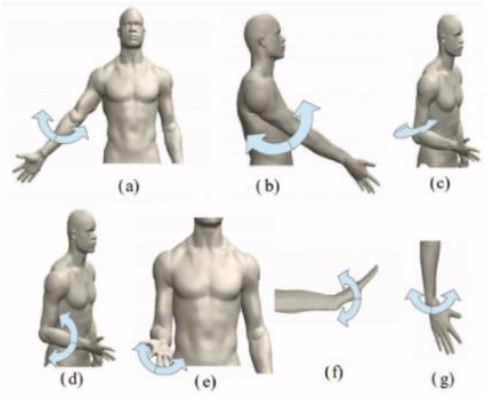


Figure 16. a) Shoulder component (Abduction and adduction). (b) Shoulder component (extension and flexion). (c) Shoulder component (rotation internally and externally). (d) Elbow component (extension and flexion). (e) Elbow/Forearm component (Pronation and supination). (f) Wrist component (extension and flexion). (g) Radial-Ulnar flexion.

Most daily activities do not require the upper limbs to operate within their full range of motion and therefore identifying the necessary range of motion for each part of the upper limb is crucial for performing these daily tasks. For example, while the typical range for elbow flexion and extension varies from 0 to 150 degrees, and for pronation and supination is from 75 to 85 degrees, a study conducted by Morrey et al. found that completing 15 common daily activities requires only from 30 to 130 degrees in elbow movement and around 50 degrees in pronation and supination. Similarly, the standard range for wrist extension reaches up to 70 degrees, with flexion between 80 and 85 degrees. However, research by Ryu et al., which examined the motion of 40 individuals during ADLs, found that most daily tasks only require about 70% of the wrist's total motion capacity. Hence, understanding the necessary range of motion for daily activities can greatly help in designing assistive exoskeletons for the elderly [14].

2.2 Mechanical Design of an Assistive Upper Limb Exoskeleton

In this section the design model of an assistive upper limb exoskeleton will be discussed. The proposed exoskeleton concept (Figure 17) is meant to be attached to a wheelchair [25].

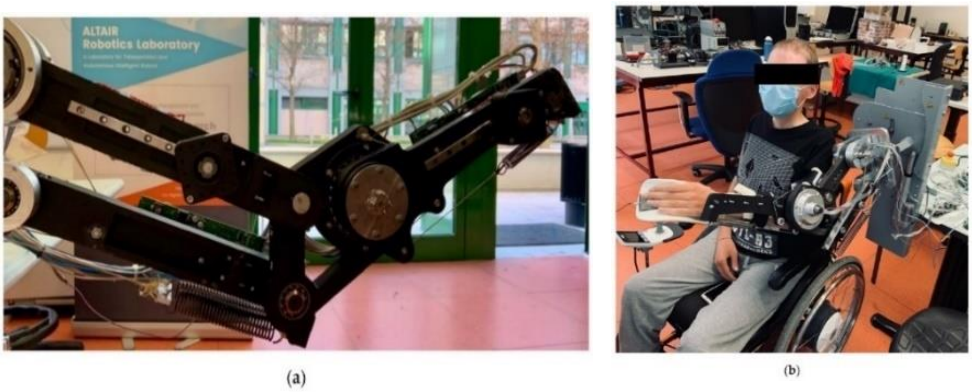


Figure 17. The novel prototype, (a) overview and (b) mounted on a patient's wheelchair.

The mechanism was created to interface with the patient's left arm and forearm. Considering that the device is intended to assist the subject in rehabilitation or in pick-and-place tasks of lifting and lowering different objects, whose trajectories mostly take place within the sagittal plane, the exoskeleton was designed to provide two-DOF on such a plane, called shoulder and elbow flexion/extension [25]. The upper-limb exoskeleton constructive design is shown in Figure 18.

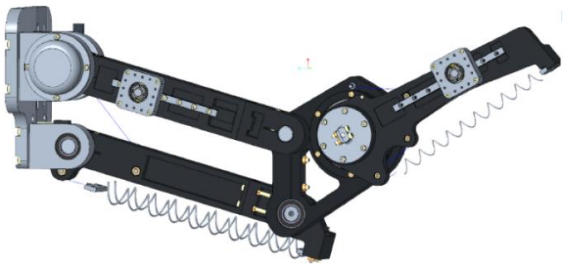


Figure 18. Exoskeleton constructive design.

The structure consists of five links labeled 1, 2, 3, 4 and 5, each one connected to the adjacent one through a revolute joint. The link 4 corresponds to the segment O_3A (see Figure 19). The constraints imposed by these revolute joints restrict the system's movement to a single plane. A parallelogram linkage, designated as O_1O_2AB and formed by links 1, 2, 3 and 4 serves as the upper arm link, while link 5 is used as the forearm link in an open chain configuration. To reduce motor demands, a spring-pulley-cable compensation system is implemented. The gravity-balancing system (GBS) is designed to account for the weight forces of both the links and the human limbs [25].

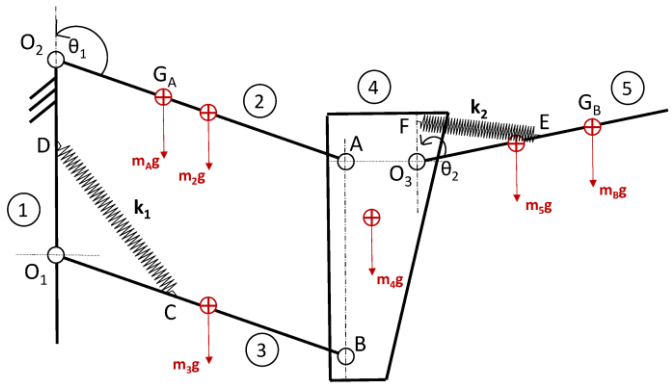


Figure 19. Parallelogram-based design.

Referring to Figure 19, m_A *and* m_B represent the masses of the forearm and upper arm respectively, while m_i (where i = 2,…,4) denote the mass of i-th link. The mass of the link 1 is not indicated because it is directly attached to the base frame. The center of mass for each link is located in its centroid (for links 2, 3 and 5, their center of mass is situated in the middle of the link, while this is not true for link 4 since it is connected to a mechanical component). The centers of mass for the limbs are positioned at points G_A *and* G_B. Two springs, CD and EF, with stiffness constants k_1 and k_2**,** are incorporated into the system. The springs are attached at points C, D, E and F. The two DOF, θ_1 and θ_2, are controlled by two identical electric motors connected to links 2 and 5, with their axes perpendicular to the motion plane, passing through points O_2 and O_3, respectively. The mechanical ranges of motion (ROM) are 165° for the shoulder and 150° for the elbow joint [25].

The final exoskeleton parameters are found in Table 2 (the mass of link 1 is not shown in the table because it is directly connected to the frame, as mentioned previously).

Link	Length (mm)	Mass (kg)
1	100	-
2	300	0.52
3	300	0.6
4	-	1.3
O_3A	80	-
5	270	0.52

Table 2. Mechanical parameters of the studied upper limb exoskeleton.

2.3 Design Difficulties

There are many design difficulties for developing a proper upper limb exoskeleton robot. Complex nature of the human upper limb makes this design very challenging. The design specifications of an upper limb exoskeleton depend on its application and on the applied limb segment [26]. The instantaneous center of rotation (COR) at the elbow and shoulder joints typically varies during joint movement and, moreover, misalignments between the joint axes of the human and the robot may occur during the motion. These misalignments can result in increased interaction forces and torques at the connection points, leading to higher cognitive loads and significant contact pressure, which can cause discomfort for the user. Various strategies are employed in upper limb exoskeleton robots to mitigate the effects of joint axis misalignment. For instance, a rehabilitation exoskeleton, called MEDARM, incorporates additional joints that allow it to adjust the shoulder's COR, considering that it shifts during motion. Therefore, a specialized linkage mechanism is designed for the shoulder joint in order to account for the COR movement. In NEUROExos, a compliant actuation system is used to minimize joint axis misalignments at the elbow joint. However, many of the complex mechanisms designed to overcome joint axis misalignment in exoskeleton robots are not practical for portable designs. Since an upper limb exoskeleton robot directly interacts with the human user, safety is crucial and for this reason mechanical singularities of the robot should be avoided in the region of motion or must be positioned in a location which is anthropometrically desirable. Some industrial designs are capable of accommodating these singularities.

2.4 Materials of Assistive Upper Limb Exoskeletons

The materials used in upper limb exoskeletons are crucial for achieving the desired balance of strength, weight and flexibility. Rigid exoskeletons are obtained from materials such as *aluminum alloys*, *titanium* and *titanium alloys* [14]. These materials can be used for assistive devices which are fixed on wheelchairs and therefore the weight of the system will not be exerted on the user. For portable exoskeletons, designed for individuals who are able to walk, composite materials such as *carbon fiber* are preferred due to their lightweight properties. However, reshaping these materials is more challenging compared to metallic options. Additionally, *soft materials* can also be utilized (*see paragraph 10.1*). These materials are chosen for their **high strength-to-weight ratio**, which ensures that the exoskeleton can provide adequate support while remaining lightweight enough for the user to ensure comfortability over extended periods of time.

a. *Aluminum alloys* are widely used due to their combination of lightness, corrosion resistance and relatively high strength. These alloys are particularly popular in joint components and structural frames, where rigidity is essential without adding unnecessary weight.

b. *Titanium* is sometimes utilized in exoskeletons requiring exceptional strength while maintaining a low weight. It is particularly beneficial in the joints or areas subjected to higher stresses due to its fatigue resistance and biocompatibility.

c. *Carbon fiber composites* offer an even lighter option compared to metals, providing strength and stiffness while reducing the overall weight. This is crucial in designs where user fatigue needs to be minimized, particularly for long rehabilitation sessions or prolonged use in industrial applications.

Chapter III - Control Strategies for Assistive Upper Limb Exoskeletons

Control strategies define the overall approach or philosophy behind the exoskeleton's assistance. These strategies describe how the exoskeleton is meant to interact with the user and the degree of assistance it provides.

There are three primary robotic-mediated control strategies for assistive devices: **assistive, corrective** and **resistive modes** [27], as shown in Figure 20.

- *Assistive Mode* means that the robot supports the weight of the impaired limb and it provides forces to finish the task. Even if the patient exerts no effort, the task can still be accomplished based on the level of support offered by the robot.
- *Corrective Mode* is applied in rehabilitation scenarios where the robot intervenes only when the patient makes incorrect movements and it will guide, when it is required, the impaired limb to execute the intended task that the patient was supposed to perform.
- *Resistive Mode,* on the other hand, involves the robot in opposing the user's movement, increasing the task's difficulty for the patient. This approach aims to encourage the patient to learn and enhance their ability to correct movement errors.

While assistance aids the patient in completing tasks, correction does not assist directly but, instead, corrects errors when they occur. Typically, assistive exoskeletons do not rely solely on one strategy; rather, they utilize a combination of these approaches. Most existing upper limb exoskeleton controllers implement a mix of assistive and corrective strategies, while resistive controls may be employed in exoskeletons to suppress tremors during exercises.

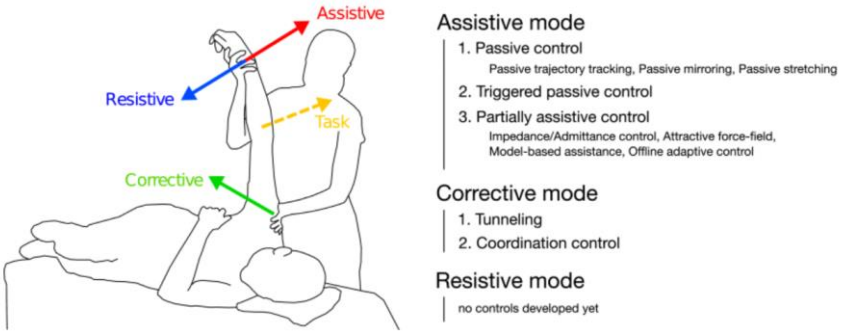


Figure 20. Three global strategies for robotic-mediated assistance and the current implementations on robotic exoskeletons.

3.1 Assistive mode

There are three types of assistive control modes: **passive control**, **triggered passive control** and **partially assistive control**. These modes are also used as a hybrid in some cases. The only difference between the passive and partially assistive mode is the participation of the patient. Therefore, if the patient is not participating, then both these modes will behave similarly without any difference.

3.1.1 Passive Control

A reference trajectory defines a suggested path that the controlled variable should follow to reach a set-point in a specified manner. The simplest approach in order to control an exoskeleton is to rigidly guide its motion along this reference trajectory using a *position feedback control*. This method is particularly useful in the early stages of neuromuscular rehabilitation, when the impaired limb is non-responsive and can only move passively, assisted by the robot. This makes the feedback control system crucial, as it must be accurately calibrated to not hurt the user when an error in its trajectory is made. Such errors may occur due to frequent muscle contractions. One solution to mitigate these errors is to implement mechanical safeguards, such as elastic straps or serial mechanical fuses. The three main passive control strategies are: **passive trajectory tracking**, **passive mirroring** and **passive stretching**.

3.1.1.1 Passive Trajectory Tracking

Passive trajectory tracking can be achieved through various techniques, with the simplest one being the use of proportional-integral-derivative (**PID**) feedback control, which regulates the position or the interaction force along a designated trajectory, and it can be applied either at the joint or at the end-effector level. Numerous methods exist for defining reference trajectories. In passive techniques, these references are typically created by recording the therapist's input on the subject limb which is attached to the exoskeleton during a teaching phase. During this phase, the robot operates in a *transparent mode* (generally achieved by incorporating a feed-forward term to compensate for the robot's gravity and dynamics) to minimize resistance in performing any type of motion. Once recorded, the exoskeleton can replay the trajectory using its feedback controller ("*teach and replay*" technique [28]).

3.1.1.2 Passive Mirroring

This type of passive control is designed for exoskeletons that support both human arms simultaneously. It involves the synchronous passive mimicking of the movement and behaviour of a healthy limb, operating in a master-slave configuration. Here, the healthy limb acts as the master while the exoskeleton serves as the slave, allowing the exoskeleton to mimic the healthy limb's motion and later operate independently in a similar manner.

3.1.1.3 Passive Stretching

In this approach, the robot passively stretches individual joints to determine their specific angle-resistance torque relationships. These measurements are then used to synchronize the passive stretching of multiple joints simultaneously. Feasibility test conducted on three stroke patients demonstrated a reduction of cross-coupled stiffness after a 40-minute stretching session.

3.1.2 Triggered Passive Control

A variation of assistive modes involves approaches where the user triggers the exoskeleton's assistance. This technique is commonly employed to integrate brain-machine interfaces (BMI) into the control loop. These methods are directly derived from assistive technologies designed for patients with non-recoverable impairments, such as tetraplegia. Once the user triggers the system, typically by selecting a target, the exoskeleton generally operates passively, following predefined trajectories.

3.1.3 Partially Assistive Control

Passive motions are found to be insufficient and limited in certain cases, as the user does not actively participate in performing the task. However, assistance is necessary at the beginning of the therapy to minimize frustrations, thus maintaining the subject's motivation, intensity of the training and confidence in using the impaired limb, avoiding negative reinforcement. If the therapy shows positive results and the patient is on the path to regain motor skills, then the control of movements must be shared between the patient and the robot. It is essential for the robot to incorporate a mechanism that allows even the slightest voluntary movement from the patient without suppressing it. There are four primary partially assistive control strategies which are: **impedance/admittance control**, **attractive force-field control**, **model-based assistance** and **offline adaptive control**.

3.1.3.1 Impedance/Admittance Control

Impedance control is a force control strategy based on a model that incorporates *position feedback*, while its dual, admittance control, is a position control method that uses *force feedback*. To enhance interaction between the robotic exoskeleton and contact forces, real-time adaptability is crucial and impedance control is considered the most effective technique for achieving this goal. Impedance control is particularly suitable for lightweight and backdrivable exoskeletons, such as cable-driven systems (*see paragraph 5.2.2*). In these systems, problems occur when compensating for gravity and friction. For exoskeletons that are not backdrivable, admittance control tends to be more appropriate, as it relies on measuring the forces at the human-robot interface to move the exoskeleton, accounting for its inertia and dynamic effects. The balance between force-tracking accuracy and the simplicity of implementation makes these two control algorithms widely used in exoskeletons for upper limb rehabilitation.

3.1.3.2 Attractive Force-Field Control

Kim et al. [29] employed an attractive force field to guide their exoskeleton during a clinical rehabilitation study involving 15 patients with motor impairments. In this study, participants were randomly divided into groups for bilateral training (BT), unilateral training (UT) or standard therapy (ST), and they completed 12 sessions of virtual reality-based exercises (using eight different games). In BT, the robotic exoskeleton provided partial assistance by mirroring the movement of the unaffected limb. In UT, the attractive force field was applied only to the end-effector, based on its position. The other joints of the robot received assistive force which was calculated using reference trajectories generated from the swivel angle estimation algorithm [30]. After the treatments, all participants demonstrated significant improvements in both fine and gross motor control across all therapies. BT resulted in better outcomes compared to UT, though this may have been due to increased robotic assistance to the affected limb. Nevertheless, robot-assisted therapy produced results similar to those achieved with ST.

3.1.3.3 Model-Based Assistance

In the study conducted by Ding et al. [31], following an offline posture assessment using motion capture, the authors developed an estimation of muscle force required to maintain a specific posture, using a musculoskeletal model of the human body. The estimated muscle force was then fed into the device's controller with the real-time force, which was measured using three EMG sensors. Then a *proportional control system* adjusted the pneumatic actuators to apply only the necessary amount of force. In [32], this approach was enhanced by implementing an optimization algorithm to minimize the activation of non-target muscles.

3.1.3.4 Offline Adaptive Control

Trial-by-trial adaptation is another approach to adjust robotic assistance according to various performance metrics. Wei et al. [33] applied an iterative learning control method, where a *feedforward assistive component* was developed using an offline adaptation. This adaptation was based on the error from the previous trial and a nonlinear function, derived from fuzzy logic, that considered tracking error statistics. The feedforward term was subsequently combined with a *PID feedback controller* at the joint level.

3.2 Corrective mode

There are two types of corrective control modes: **tunneling** and **coordination control**.

3.2.1 Tunneling

Tunneling involves creating virtual channels for the exoskeleton's end-effector or for the joints in which the user moves. If the user leaves from these channels, the input control system guides the patient back as if a spring-like impedance was connected from the limb to the center of the virtual channel. Additionally, to prevent the user from getting stuck during its motion, a supportive force is typically applied toward the channel. Therefore, tunneling techniques can be regarded as a form of impedance control with a designated no-action zone; the supportive force field can be seen as an assistive term, while the spring-damper-like force, pulling toward the channel's center, acts as a corrective term.

3.2.2 Coordination Control

In corrective modes, the control law is typically designed to manage coordination during movement, therefore it regulates joint positions and/or velocities. Coordinating the positions or speeds of different joints helps address the issue of reference time dependence, which continues to be a limiting factor for assistive controllers.

Chapter IV - Control Systems for Assistive Upper Limb Exoskeletons

Control systems of exoskeleton robots significantly differ from those of traditional robots. This difference arises because, in these systems, the human operator is not just a supervisor or the commander of the system but it is also an integral part of the control loop. As a result, this concept is often referred as the “man-in-the-loop” approach. The human operator primarily makes decisions, while the exoskeleton carries out the tasks. However, there is a continuous bilateral feedback between the operator and the exoskeleton robot. Consequently, the intelligence of the exoskeleton system is enhanced, while the operator's power is simultaneously augmented. A key factor in controlling an exoskeleton robot, particularly a power-assist exoskeleton, is ensuring it responds to the user's intended motions. This is especially crucial for individuals with physical weakness, who *may struggle to perform daily activities independently*.

4.1 Control Methodologies

Control methods focus on the type of signals used to control the upper limb exoskeleton and how these signals are processed. These methods determine how the exoskeleton interprets user intentions or reacts to feedback information and adjust its movement accordingly. In general, exoskeleton robots consist of two types of controllers which work in parallel to each other and they are called: **exoskeleton controller** and **human brain**. Selecting the most effective control method is challenging because predicting patients’ movements is difficult and therefore optimizing such methods becomes even more complicated. Control methods of upper-limb exoskeleton robots can be classified in several ways**: based on input signals to the controller**, **based on controller configuration** (architecture), **based on output from the controller** and other ways [34]. Figure 21 shows the ways of classification of control methodologies.

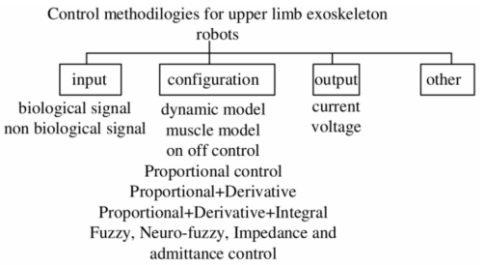


Figure 21. Classification of control methodologies for upper limb exoskeleton robots.

Control methods of recent upper limb exoskeleton robots are compared considering few points: their application, special features and applied robot. **In this section, only control methods based on input signals will be discussed in depth**, since controller inputs are very significant for the performance of the system.

4.1.1 Control Methods based on Input Signals to the Controller

Based on input to the controller, the control methods can be categorized into: **control methods based on human biological signal**, **control methods based on non-biological signal** and **platform independent control methods** [34]. Figure 22 illustrates the classification of control methods for upper limb exoskeleton robot *based on input signals*.

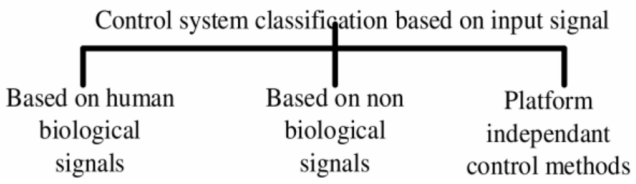


Figure 22. Classification of control methodologies based on input signals.

4.1.1.1 Control methods based on Human Biological Signals

The main goal of these control methods is to manage successfully the system under the effect of various disturbances from the external environment. In the context of exoskeleton robots, traditional control concepts have been expanded to incorporate human motion intentions. This integration is a notable aspect of exoskeleton control methods, with one type of human biological signal, known as electromyography (EMG), being successfully utilized in several designs. Most EMG-based control strategies for exoskeleton robots operate on a binary (on-off) nature. EMG signals indicate the firing rate of motor neurons, which is proportional to the muscle activity levels. Therefore, these signals serve as a direct indicator of human motion intention. However, in practice, measuring motor neuron activity can be challenging. EMG signals can be obtained in two ways: using **surface EMG** (**sEMG**) **electrodes** or **intramuscular EMG electrodes**. The latter generally provides a clearer pattern of muscle activation than sEMG electrodes. Nevertheless, implementing EMG-based control systems presents difficulties due to various factors. In fact, all EMG-related parameters are subject-dependent and can change from day to day due to varying conditions of the skin and body size of the patient. Incorporating muscle models (myoprocessors) and exploiting the natural electromechanical time delay in human neuromusculoskeletal physiology, this control system can anticipate the operator's intentions before movement begins.

4.1.1.2 Control methods based on Non-Biological Signals

These methods rely on external sensors like force sensors or position sensors and they are based on environmental data to guide the movements of the exoskeleton. For instance, force or torque sensors can be used to adjust the exoskeleton's response based on user interaction with the environment, regardless of biological inputs. These approaches are often applied when EMG signals are insufficient or unreliable.

4.1.1.3 Platform Independent control methods

These control methods refer to techniques that do not depend on a specific hardware or software platform, allowing for broader flexibility across different devices. The goal is often to maintain high adaptability for different users, including those with varying physical capabilities. Such control systems often focus on integrating human-robot interactions through various techniques like impedance or adaptive control.

Chapter V - Actuation, Power Transmission and Sensing Methods

5.1 Actuation

Exoskeletons can be classified in two categories: **active** and **passive**. *Active* exoskeletons utilize external actuators like electric motors, hydraulic actuators or pneumatic muscles to boost the user's strength and provide additional energy. On the other hand, *passive* exoskeletons do not employ actuators. Instead, they use components such as springs and dampers to store and release the energy generated by the user's movements, aiding in maintaining a posture or in supporting motion at specific joints. There are five primary modes of actuation: **electric motors**, **hydraulic/pneumatic actuators**, **series elastic actuators** (**SEAs**) and **pneumatic artificial muscle actuators** (**PAMAs**) [11, 35]. The choice of the actuators in exoskeleton design is crucial, as it can increase the system's overall weight, and there is the need of an appropriate torque in order to drive correctly the upper limb exoskeleton. However, the selection also depends on the specific application for which the exoskeleton is intended. Thus, it is recommended to select an actuator which can have a **high power-to-weight ratio** and can produce high torque with precise movement.

5.1.1 Electric Motors

Electric actuators are predominantly used because they efficiently store electrical energy and offer excellent controllability through advanced motion control techniques. Electric motors are used by many upper limb exoskeletons to achieve high control precision and quick response [36]. However, their typically low power-to-mass ratio presents a problem for upper extremity exoskeletons, particularly for those with a high degree of freedom and complexity, which require additional actuators. This often requires a trade-off between power and portability. Another disadvantage is that electric motors tend to be expensive.

5.1.2 Pneumatic Actuators

Pneumatic actuators operate using compressed air, necessitating the use of specialized compressors or air containers for power. These actuators typically have lower impedance and are lighter than electric motors. They require minimal maintenance and can be stopped under load without compromising patient safety, but they are less prone to hazardous and humid environments. However, the primary disadvantage is their limited precision and accuracy. For this reason the number of exoskeletons utilizing pneumatic actuation is relatively small. Since a pneumatic system needs air pressure, exoskeleton robots equipped with pneumatic actuators are often designed as stationary platforms, which restricts their operational range. In some cases, a compressor must be installed for user convenience.

5.1.3 Hydraulic Actuators

Hydraulic actuators are powered by hydraulic pressure and can generate greater torque compared to electric and pneumatic ones. However, the system is relatively complex, as it requires careful maintenance of pressurized oil to avoid leaks. These actuators also demand additional space to accommodate oil pipes and conduits. Commercially available hydraulic systems tend to be heavy, therefore only specially designed hydraulic systems are required for rehabilitation robotics. The inclusion of extra components such as fluid reservoirs, motors, pumps, valves, heat exchangers and noise-reduction equipments make this type of actuator

more suitable for human capability augmentation rather than for robotic physical therapy aimed at helping elderly individuals and patients.

5.1.4 Series Elastic Actuators (SEAs)

SEAs reduce inertia and improve the user interface, allowing for more accurate and stable force control, which enhances safety. However, this is not always guaranteed. The elastic energy stored in the spring can be suddenly released during an impact or improper handling of the device, potentially leading to unexpected reaction forces [11].

5.1.5 Pneumatic Artificial Muscle Actuators (PAMAs)

A pneumatic artificial muscle actuator (PAMA) is made up of a flexible inflatable membrane enclosed in a fibrous material, with one end connected to the load and the other end attached to the gas or air valve. When air is introduced into the membrane, it starts to expand radially while contracting along its axis, generating force in the axial direction. PAMAs replicate the nonlinear behaviour of biological muscles, making them challenging to control. However, they offer a significant advantage with a power-to-weight and power-to-volume ratio five times higher than that of electric actuators. Despite these benefits, PAMAs lack durability under high-performance demands [35].

5.2 Power Transmission

There are different methods to transmit power along the upper limb exoskeleton. The main ways for power transmission are: **direct drive, gear or linkage** and **cable driven** [36].

5.2.1 Direct drive, gear or linkage

To directly achieve power transmission from a rotary or linear actuator, one may choose to locate the actuator right next to the anatomical arm joint and connect the links to the human arm's linkages. This approach works effectively for elbow flexion/extension, which is often the first joint addressed in exoskeleton prototypes. However, as the DOF increases, such as with the shoulder, more sophisticated mechanisms are required to allow movement while avoiding bulky designs and potential singularities, ensuring the maximum reachable workspace (both in task and joint space). Therefore, straightforward solutions like: direct drive, gears or linkages can be introduced but, in most cases, they significantly increase the weight of the system and this may require active gravity compensation in the controller and can also reduce safety in pHRI.

5.2.2 Cable-driven

This transmission approach allows for a lightweight design, since all the actuators, even if they are bulky or heavy, can be placed at the back or end of the exoskeleton system. This helps to reduce the size and weight of the components that are in direct contact with the user's upper limb. However, the control could be an issue since the torque output might be compromised and, in this case, a stronger cable is used to compensate.

5.3 Sensing and Estimation

Sensing and estimation are key components in designing an upper limb robotic exoskeleton. *Proprioceptive sensors*, used at a lower level, measure physical states of the exoskeleton such as joint position, speed and motor torque through feedback control. These sensors provide accurate property estimation for nearly all electrical actuators. At a higher level, *exteroceptive sensors*, like force-sensitive resistors (FSR), load cells, electromyography (**EMG**) sensors and electroencephalography (**EEG**) sensors, define the task-oriented interpretation of sensor data and allow the integration of information across space and time [11]. FSR sensors and load cells are commonly used to measure the interaction pressure across the contact area between the upper limb and the exoskeleton. Besides, several studies have introduced the use of neural and muscular sensing for exoskeleton control, highlighting its importance. For instance, HAL is the only commercially available exoskeleton that employs EMG data as a control input, but numerous research prototypes have integrated EMG sensors data as the reference signal. While brain and neural signals have not yet been adopted in commercial products, researchers are investigating the use of EEG sensors, particularly for high-level spinal cord injury patients.

5.3.1 Classification of the Sensors used in Upper Limb Robotic Exoskeletons

The movement of an upper limb exoskeleton can be divided into several modes [36]:

- *Resistive trainer*: applies resistance to strengthen muscles.
- *Orthosis*: corrects abnormal movements by exerting force in different directions.
- *Pure follower*: detects the user's intent and follows their movements with minimal resistance.
- *Assistive device*: provides supportive force to complete tasks.
- *Predefined trajectory*: controls the user's motion through position control.
- *Capability augmentation*: compensates for the arm's weight and any additional load.

These movements require sensors to guide the exoskeleton effectively. The main industrial sensors used in upper limb exoskeletons are: **position sensors, force/torque sensors, Inertial Measurement Unit (IMU) sensors**, **EMG sensors** and **EEG sensors.** The first three categories are usually exploited in order to implement control methods based on non-biological signals, while the last two categories are used to implement control methods based on human biological signals. Integrating various sensing techniques, through sensor fusion, could improve both the stability and safety of the system's intention detection.

5.3.1.1 Position sensors

A crucial factor in understanding the spatial arrangement of an exoskeleton and in examining its kinematics and dynamics is its position. The position is usually evaluated directly or indirectly through different types of sensors, including encoders, potentiometers, flex sensors, and transducers. When choosing a specific type of position sensor, considerations such as the sensor's footprint, measurement precision, data transmission speed, reliability and cost often play a significant role.

5.3.1.2 Force/Torque sensors

When dealing with features that require force data, like providing assistance, force and torque sensors are essential for indicating to the exoskeleton when to start and stop the movement. These sensors are also critical in haptic applications, including virtual reality (VR) teleoperation and rehabilitation. In certain situations, extra force and torque sensors may be incorporated to enhance the system's safety.

5.3.1.3 IMU sensors

IMU sensors effectively monitor motion. These sensors are made up of a gyroscope and an accelerometer, used to detect the deflection, distance, angular velocity and rotation angle of an object.

5.3.1.4 EMG sensors

Electromyogram (EMG), typically measured in a non-invasive way by placing electrodes on the arm's surface (sEMG), can identify upcoming muscle activity and help the exoskeleton to determine which joints to move in response to the user's arm motions. When the signals are accurately filtered and analyzed, sEMG can effectively control the exoskeleton’s movement, theoretically eliminating the need for physical force interaction between the arm and the device. However, several practical challenges complicate the implementation of this sensing method: the setup and calibration processes can be time-consuming, the signals are often noisy and high-degree-of-freedom exoskeletons may necessitate intricate and costly multichannel sEMG systems.

5.3.1.5 EEG sensors

The EEG sensor technology is widely recognized as a brain-machine interface (BMI). At a particular moment in time, the EEG reflects the full spectrum of brain activity visible at the electrode's location, making it one of the most practical and realistic methods for a non-invasive brain-machine interfacing. Like sEMG, EEG provides comparable advantages and can be utilized for stroke patients who have lost the ability to move their arms. However, EEG signals can be as noisy as those from sEMG, necessitating substantial processing efforts.

A Comparison between different Assistive Exoskeletons in terms of Sensing Methods and Applications

	MyoPro by Myomo	***HAL by Cyberdyne***	***AXO-SUIT (Upper Limb Component):***	***Power Jacket by Panasonic***
SENSORS	Electromyography (EMG) sensors	A combination of bioelectric sensors for neural signal detection, including sEMG and EEG sensors	Force sensors distributed along the arms and joints	Pneumatic systems with air pressure sensors and embedded EMG sensors
ADVANTAGES	EMG sensors detect electrical signals from muscle contractions, allowing for intuitive control. This is particularly beneficial for people with partial muscle function, as the sensors help in tailoring the assistance based on real-time muscle engagement	This exoskeleton offers unique brain-driven control, which allows for highly responsive movement assistance. It can detect weak neural signals, making it adaptable even for users with more severe motor impairments	Force sensors allow for a tuned force assistance, adapting to the user's physical output. This makes the AXO-SUIT effective in scenarios where gradual assistance is needed, such as lifting or reaching light objects	The pneumatic approach allows for soft and gradual movements, reducing the risk of sudden jerks, which is advantageous for fragile users or those with spasticity. The EMG sensors further refine the control, making movement smoother and more responsive to muscle signals
DISADVANTAGES	EMG-based control can be less effective for users with minimal or inconsistent muscle activity, such as advanced paralysis or degenerative muscular disorders. Muscle fatigue or skin interferences can also affect signal's quality, leading to potential inaccuracies	Neural sensors, such as EEG, require careful placement and calibration, which may increase setup time and complexity. These sensors are also very sensitive and may not perform optimally in environments with high electrical noise	Force sensors can add bulk to the exoskeleton, impacting user comfort and natural movement. Additionally, force-sensing accuracy can be affected by sensor misalignment or calibration drift	Pneumatic systems are generally larger and may limit portability. Maintenance of air compressors or pneumatic lines can also be cumbersome, especially in home settings
APPLICATIONS	The MyoPro is suitable for tasks like eating, grooming, using a computer, aiding veterans and others people with partial upper-limb impairment [37]	HAL is used extensively in rehabilitation centers to help users regain basic motor skills through neural engagement and is suitable for assisting patients with stroke or neurological impairments [23]	AXO-SUIT is designed for older adults or those with mild strength limitations. It helps in ADLs by offering incremental support during simple tasks [38]	Power Jacket is primarily aimed at stroke survivors and elderly individuals. This exoskeleton provides support in daily tasks such as dressing or using utensils, helping to counterbalance limb weakness [14]

Table 3.

CHAPTER VI- Possible Challenges in Upper Limb Exoskeleton Development

The exoskeleton technology has demonstrated its significant impact across various fields of application, as explained previously. However, numerous challenges and open research questions still need to be addressed.

6.1 Kinematic compatibility

The design of an exoskeleton must be kinematically compatible with the user's varying anthropometric parameters. Achieving proper alignment between the exoskeleton and the anatomical joints of the wearer remains a challenge. In particular, the exoskeleton mechanism, that supports the shoulder and wrist joints, needs to function as a kinematic spherical joint to replicate natural shoulder and wrist movements, overcoming misalignment issues and ensuring biomechanical compatibility of the design [11].

6.2 Workspace limitation

Wearable exoskeletons are limited in providing the same range of motion as the human upper-limb torso. For instance, studies [39,40] have introduced new mechanisms to support three-DOF shoulder glenohumeral (GH) movements, and to also maintain the instantaneous center of rotation (COR) during various activities of daily living (ADLs). These mechanisms generally reduce the actual workspace of the upper limb, but this can be considered acceptable in some applications. Consequently, multiple studies have focused on overcoming this workspace limitation by incorporating shoulder girdle movements, at the cost of more complex and heavier designs. It has been observed that the required workspace and range of motion (ROM) vary depending on the specific application. For this purpose, it is essential to evaluate the combined ROM of both the human arm and the exoskeleton to achieve the desired workspace and this can be done by analyzing the kinematic and dynamic models of both systems. However, achieving a wide ROM generally comes at the expense of increased complexity and weight in the mechanism.

6.3 Singularity problem of mechanical system

Singularity problems hinder the smooth interaction between the exoskeleton and the human upper limb. This is is a significant issue that occurs when two joints of the exoskeleton become aligned, causing loss of degrees of freedom of the system and requiring infinite torque to move it out of the singular configuration. There are two ways to address this issue. One method is to account for these potential situations in the exoskeleton's control strategy. The other is to overcome it through design by introducing mechanical constraints that prevent such alignments [11].

6.4 Discomfort and Misalignment

The comfortability level of wearable exoskeletons needs to be measured, but no standards currently exist, which is a major challenge. Ideally, the exoskeleton should be perfectly aligned with the body’s physiological joints. However, achieving this alignment is particularly difficult for the shoulder, wrist and thumb/finger joints. Unlike conventional mechanical joints, physiological joints are highly complex and their rotational axes shift during movement. Moreover, the connections between the exoskeleton and human limbs are

not inflexible, meaning slippage between them can occur while performing the task. This slippage leads to misalignment between the exoskeleton and the physiological joints, causing unwanted forces and torques at the human joints. Different studies have already tried to overcome the joint misalignment issues by considering the complex human biomechanics.

6.5 Human-Robot interaction

A crucial aspect of controlling active exoskeletons is determining the human intent. This can be achieved by analyzing the interactions between the user and the exoskeleton. Decision-making is performed through a cognitive process that relies on sensory input from the surrounding environment, such as visual, tactile and auditory signals. The musculoskeletal system cognitive process is influenced by the robotic exoskeleton while driving human joints. This is referred to a concept known as **human-robot interaction** (**HRI**) [41]. In terms of control, the user and exoskeleton together form a closed-loop system based on human-robot cooperation.

Depending on how HRI is measured, exoskeleton control can either follow a **cognitive human-robot interaction** (**cHRI**) or a **physical human-robot interaction** (**pHRI**). In cHRI systems, electric signals from the central nervous system to the musculoskeletal system are used as inputs for controlling the exoskeleton. This allows for identifying the user's intent before any actual motion occurs, predicting the necessary torque, velocity or position required for joint motion. In contrast, pHRI-based control relies on measuring forces or positional changes caused by musculoskeletal movements in order to direct the exoskeleton.

Both pHRI and cHRI can be directly measured. For instance, data from EMG or FSR sensors can be processed to classify upper limb movements and then can be used to control the robot. However, accurate measurement of cognitive or physical interactions depends on the user's condition and on the choice of the muscle group involved.

CHAPTER VII – Evaluation methods, clinical overview and market integration of upper limb assistive exoskeletons

The performance evaluation of upper limb exoskeletons is a critical step in understanding their effectiveness and optimizing their functionality. This involves assessing various metrics such as ergonomics, adaptability and user satisfaction. In parallel, the availability of different assistive exoskeletons designs reveals a growing diversity of solutions tailored to support individuals with mobility impairments or physical problems. Similarly, the integration of these devices into the market offers significant advantages, including improved accessibility and the potential to alleviate the socioeconomic burden of physical disabilities. In the following paragraphs, the **methods for assessing the performance of upper limb exoskeletons** will be presented. Additionally, a **clinical review of various upper limb assistive exoskeletons** is provided and the discussion will also be extended to their **economic implications** and **market adoption**.

7.1 Performance Evaluation Methods

Most mechanical designs of upper limb exoskeleton robots have been evaluated by considering the specific functions of the robot [42]. For example, Kiguchi et al. [43] assessed the mechanical design of an upper limb exoskeleton by analyzing its *power-assist capability*. A commonly used quantitative measure to evaluate system performance is the **bandwidth.** Systems having a higher bandwidth are controllable under higher frequency command signals [44]. Bandwidth reflects how well trade-offs between weight and stiffness of upper limb exoskeletons are managed. To ensure high performance, the exoskeleton's lowest natural frequency should be above the highest frequency command signal generated by the human. In many cases, *proportional-derivative* (**PD**) control methods have been used to evaluate the mechanical design of exoskeleton robots. Kawasaki et al. [45], for example, measured the frequency characteristics of a robot using PD control to study its behavior. Additionally, joint angle responses were monitored to examine tracking properties. A sinusoidal signal with an amplitude of 1 degree was applied to measure frequency characteristics as a linear system. To dissipate the gravity effect, each joint axis was aligned in the direction of gravity. Moreover, a dummy hand, having flexible finger joints, was attached to the robot in order to maintain a closed loop system avoiding free motion by a passive joint.

7.2 Clinical Reviews of Various Assistive Robotic Exoskeletons

As already explained, spinal cord injuries (SCI), strokes and ageing are the leading causes driving the need for upper limb assistive systems among patients. Although robotic rehabilitation is a relatively new field, it has gained significant attention worldwide due to its greater effectiveness compared to traditional physiotherapy. Most robotic rehabilitation systems allow for precise control of articular joints by guiding the patient’s limb based on human anatomy. One example of such a system is the Arm Light Exoskeleton (ALEx), as depicted in Figure 22.

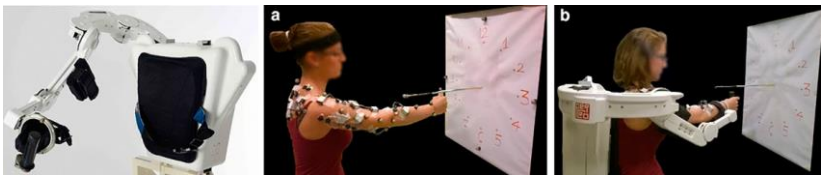


Figure 22. Arm Light Exoskeleton (ALEx), experimental setup in (a) and (b).

After examining 16 upper limb muscles and tracking movement in six patients, it was found that ALEx reduced the shoulder abductor muscle activity and increased the elbow flexor activity. However, the ROM provided by the exoskeleton was reduced compared to the body's natural movements [46].

Signal processing has played a key role in robotic rehabilitation, as demonstrated by F. Xiao and colleagues in China. They evaluated the assist capabilities of a 7-DOF cable-driven exoskeleton by analyzing sEMG signals from key muscle groups involved in upper limb movement. Their findings suggested that post-stroke patients could undergo effective rehabilitation using the exoskeleton [47]. While these robotic rehabilitation systems represent significant advancements, they are still facing challenges, such as limited portability, ease of use and high manufacturing costs. To address these issues, Chinese researchers J. Huang et al. developed a more affordable, portable and in-home rehabilitation system called RUPERT. This 5-DOF exoskeleton is powered by pneumatic muscles, making it lighter than many other exoskeletons. It assists patients with daily activities both in clinical and at home through a virtual environment. The system has received positive feedback from patients and it shows promising future potential [48].

Another system that incorporates virtual environments was developed by W. Qingcong et al. They used a novel *patient-active admittance control strategy* for users with arm motor impairments, helping them perform daily tasks in a virtual setting.

Moreover, a lightweight exoskeleton was developed by A. Frisoli et al. in Italy. This device used *impedance control* to assist patients with passive and active reaching exercises, adapting to each patient’s individual needs and helping them to regain motor functions [49].

7.3 Market Integration and Economic Impact

The socioeconomic impact of upper limb exoskeletons is becoming increasingly significant in both the **rehabilitation** and **industrial markets**. In the rehabilitation sector, upper limb robotic exoskeletons are revolutionizing post-stroke recovery helping individuals to regain motor function after neurological injuries. In fact, exoskeletons improve outcomes for patients with motor impairments, leading to faster recovery times and reducing the need for long-term care. Studies have shown that robot-assisted rehabilitation increases efficiency and can offer more standardized treatment than manual therapies, resulting in potential healthcare system savings. While, on the industrial side, exoskeletons are being employed to reduce work-related musculoskeletal disorders by providing physical support during repetitive or strenuous tasks, particularly in sectors like automotive and manufacturing. This can lead to increased worker productivity, lower rates of workplace injuries and reduced compensation costs for companies. The use of these technologies helps mitigate the physical strain on workers and extends their careers, which translates to substantial long-term cost benefits for industries reliant on manual labor. Both markets, that are still in the development phase, are growing rapidly, with increased investments in wearable robotics. This growth is driven primarily by advancements in healthcare, rehabilitation and industrial sectors. The market is expected to continue growing, especially in regions like Asia-Pacific [50]. However, challenges such as affordability, widespread adoption and ensuring user comfort continue to influence their socioeconomic reach. Some reports highlight the need for further research to evaluate the long-term cost-effectiveness of these devices, particularly in non-rehabilitative settings like factories.

CHAPTER VIII - Discussion and Future Prospective

The majority of elderly individuals face limited mobility in their upper limbs, which brings challenges in performing daily activities such as dressing, eating and general self-care [14]. To improve the quality of life for older population, wearable robotic solutions have been developed to help them maintaining independence. Due to age-related muscle decline, upper limb exoskeletons can be helpful in performing various tasks like moving, carrying and holding different things. Moreover, since exoskeleton robot systems closely interact with the human body, safety must be prioritized. Safety measures are integrated into three main areas: **mechanical**, **electrical** and **software systems** [44]. Mechanically, physical stops prevent hyperextension or hyperflexion of joints. In terms of electrical design, there are three emergency shutoff switches: an enable button that turns off the motor power when it is released, a large emergency button for complete power shutoff by the observer and a similar e-stop foot switch for the user. Ideally, these safety measures would go unused as a result of adequate safeguards at the software level. Position sensors (potentiometers, shaft encoders etc.) are placed at both ends of the power train to monitor joint movement and motor position. This redundancy in position sensing allows the software to continuously evaluate the integrity of power transmission. If any slip occurs between the motor and the end effector, a discrepancy in position is detected, triggering an immediate system shutdown. Additionally, any shutdown request from a safety subsystem will activate brakes in the shoulder and elbow joints. To further enhance safety, software limits are also applied to regulate motor currents, which control the motor's torque output. Further research is needed to develop new intelligent safety systems. One potential approach involves using the **electrooculogram** (**EOG**) [51], consisting in a *human-biological signal* generated by eye movements, as a feedback mechanism. In this case when users feel any unsatisfactory movement of the exoskeleton, particular eye ball movement can be used as feedback signal in order to switch off the function of robot. As this method is directly connected with human functions, it offers optimal protection. Future developments could also focus on adapting exoskeleton control systems to account for environmental factors, such as microclimate conditions surrounding the user.

This paper has presented a comprehensive overview of upper limb robotic exoskeletons, in particular those that can help older group of people or post-disease patients in performing daily activities. Moreover, the technological insight in terms of DOF, actuation type, embedded sensors, control architecture and type of assistive motions has been discussed.

8.1 Key Innovations and Developments

In recent years, exoskeleton technology has evolved by integrating advanced materials, customized actuation systems and new control strategies. Most commercial assistive exoskeletons, however, remain cumbersome and heavy, making them impractical for usage during daily activities at home. Taking into consideration that the primary users of such devices often have limited physical strength, heavy exoskeletons can cause faster muscle fatigue. To alleviate the load on the upper limbs, placing batteries and actuators proximally to the body is essential. This makes remote actuation crucial, which can be achieved through mechanisms like *Bowden cables*, or *hydraulic* and *pneumatic systems*.

Recently, significant attention has been given to the development of **Soft Exoskeletons** [52], which utilize innovative textiles to enhance comfort and usability. These designs significantly improve upper limb mobility, especially for tasks like grasping and holding. Compared to traditional rigid exoskeletons, soft models offer several advantages, such as being lightweight,

easier to wear and more interactive with clothes, making it less visible. They can also be made in various sizes to accommodate different body shapes and their flexible structure means they don't restrict joint movement, allowing for more natural body motions. The actuation systems in soft exoskeletons are inspired by biological muscles, with both cable-driven electromechanical actuators and pneumatic actuators proving positive impact in assisting upper limb movements. Because they are lighter and more compact, they require less energy to operate.

Despite these advantages, soft exoskeletons have some drawbacks. The absence of a rigid frame can complicate the placement of batteries and actuators. Moreover, developing control systems tailored to individual users' needs is a significant challenge. Additionally, current sensing mechanisms for soft exoskeletons are not yet advanced enough to accurately determine the user intent or the required level of assistance, which can lead to robot's delays in response time and unwanted movements, such as collisions. This highlights the need for more sophisticated sensing systems and control algorithms capable of real-time adjustments to optimize the support provided by soft exoskeletons.

Furthermore, the integration of artificial intelligence (AI) has greatly improved the adaptability and personalization of exoskeletons, enabling more effective rehabilitation outcomes through intelligent control strategies based on multiple sensors. This is expected to further enhance their utility and acceptance, particularly among the elderly population.

8.2 Barriers to Progress

Despite recent technological advancements, several challenges persist. One of the key challenges is **improving the clinical validation** of these devices to ensure that they can consistently deliver reliable outcomes across a wide range of patient populations. While AI integration and sensor data fusion have advanced control strategies, there is still a demand for more intuitive systems that require minimal user interaction. Another ongoing challenge, also mentioned previously, is the **weight and size** of upper limb exoskeletons. Although designs have become more compact and lighter, further reductions are essential to improve comfort, particularly for wearable systems intended for daily use. This is crucial for ensuring user compliance in rehabilitation, where long-term use of exoskeletons is vital for recovery.

CHAPTER IX – Conclusions

Upper limb exoskeleton robots play an important role in the field of rehabilitation, assistance and human power augmentation. So far, several exoskeleton robots have been developed for various purposes and they represent a transformative advancement for both healthcare and industry.

The mechanical design of these systems is still heterogeneous due to their wide range of applications. As a result, there are no standardized guidelines for designing and evaluating the performance of upper limb robotic exoskeletons. Further research is necessary to develop new mechanisms that consider the complexity of human biomechanics, particularly in the shoulder and wrist joints. Additionally, the physical interaction between humans and robots is crucial and it has not been addressed satisfactorily in most of the recently commercialized systems and therefore it is essential to quantify the impact of robotic exoskeletons on the human upper limb, considering comfort and compliant interaction. However, achieving more seamless HRI, while ensuring safety and minimizing fatigue, remains a key area of focus.

The demand for assistive wearable robots, particularly for elderly people, is on the rise [53]. With the global population aging rapidly, there is an urgent need for affordable and scalable solutions that allow older individuals to maintain independence through assistance in activities of daily living. Exoskeletons designed for these tasks need to be integrated with daily environments, while being affordable enough for widespread adoption, which calls for collaboration between governments, healthcare providers and technology sectors. Investments in this field promise substantial socio-economic benefits, reducing healthcare costs by facilitating at-home care, lowering physical demands on caregivers and increasing productivity in industrial applications where such devices can augment human capabilities. Thus, upper limb exoskeletons hold the potential to revolutionize rehabilitation and assistive technologies, marking a significant step toward more inclusive and supportive societies.

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